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Design methods of the ABB ALSTOM POWER gas turbine dry low emission combustion system

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Abstract: This paper describes the conceptual design and aerothermal approach of the ABB ALSTOM POWER UK Limited (ABB ALSTOM POWER) dry low emission (DLE) combustion system directed towards all current and future power ranges. A set of parameters was employed as a common criterion to design the company's combustion system across the product range. The parameters made it possible to scale the combustion system to operate at any required firing temperature without the penalty of achieving the low emission targets. The logic and sensible air mass mapping and testing management based on a basic knowledge of the aerothermodynamics of the combustion system and its integral parts produced reliable operation and engine results.

The combustion system aerothermodynamic design considers the constraints of maintaining combustor specifications while introducing a reliable ultra-low NO_x concept with minimal structural change to engine design. Furthermore, the ignition loop, piloting and stability margins were issues continuously assessed through a defined application programme to examine their impact on engine operability at various load conditions.

A main radial swirler injector is utilized for both gas and liquid to produce a fuel lean mixture for combustion at full-load operating conditions, while a pilot fuel injector guarantees a local fuel-rich zone to sustain stability at low power conditions. The carefully planned switch-over between the pilot and main swirler provides smooth operation and low emission across the load range conditions.

The company combustion system relies upon the principle of aerodynamic variable mixing with a single vortex generator burner. The partial premixing process takes place at the radial inflow swirler slots before the fuel–air mixture is introduced into a prechamber and thereafter into the combustion chamber. For vane passage injection the liquid fuel atomization should be at an optimum and the vane passage and prechamber length set to provide the vaporization period. The impingement cooling scheme aided by a thin layer of thermal barrier coating (TBC) achieved an excellent combustor wall temperature at all operating conditions and across the DLE engines product range. The scheme made it possible to direct a large amount of air towards the swirler to achieve the lean premix combustion.

Substantial CO emission reduction at part load was achieved by variable guide vanes (VGV) modulation for the single-shaft engines. This action inhibits NO formation without slowing the CO oxidation process to a frozen condition. The variable mixing regime approach and accurate air management has demonstrated an ultra-low NO_x emission of 12–15 and 32–35 ppmv (corrected to 15% O_2) at the full-load condition while operating on gas and distillate-2 fuel respectively. Moreover, the system exhibited low CO and dynamics oscillation capability for both gas and liquid fuel operation.

At the present time, the one-dimensional analysis has been proved to be a reliable tool for combustion system designers to produce a preliminary design configuration. The particular choice for a commercial computational fluid dynamics (CFD) code to satisfy various criteria of the swirling combusting flow is proving to be difficult due to constraints regarding solver and/or methods limitation.

The results obtained from the high-pressure air facilities, engine beds and field trials represent a true technological achievement through collective engineering design efforts towards emission and pollution control for small- and medium-range gas turbine engines.

Keywords: design, aerothermodynamics, combustion, DLE, gas turbine, premixing, emission, NO_x

NOTATION

A_e effective area (m^2)

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B	stabilizer blockage (%)
c_1, c_2, c_3	constants determined experimentally
C	constant
C_D	discharge coefficient (dimensionless)
C_f	constant function of fuel property and injector geometry
d	pre-chamber internal diameter (m)
D	diameter (m)
G_c	cross coolant air mass flowrate (kg/m^2 bara)
G_j	coolant jet air mass flowrate (kg/m^2 bara)
m_{air}	air mass flowrate (kg/s)
M	total air mass flowrate (kg/s)
M_1	Mach number (dimensionless)
P_t	total pressure (Pa)
P_2	compressor delivery pressure (Pa)
ΔP	pressure drop (Pa)
$\Delta P/P$	impingement liner pressure loss (%)
s	mixing parameter (%)
S_g	geometrical swirl number (dimensionless)
T_{flame}	maximum flame or reaction temperature (K)
T_1	entry temperature (K)
T_2	compressor delivery temperature (K)
V_f,V_a	mean velocity of the fuel and air respectively (m/s)
V_{max}	maximum mean axial velocity (m/s)
V_{ref}	reference velocity (m/s)
x	pitch diameter in streamwise direction (m)
y	pitch diameter in the spanwise direction (m)
Y_{max}	maximum jet penetration (m)
z	gap between the impingement skin and flame tube (m)
θ	swirl angle (dimensionless)
ρ_f, ρ_a	density of the fuel and air respectively (kg/m^3)
σ	standard deviation (dimensionless)
ϕ	equivalence ratio (dimensionless)

Abbreviations

AFR	air–fuel ratio (by mass)
CFD	computational fluid dynamics
CO	carbon monoxide (ppmv corrected to 15% O_2)
CPU	central processing unit
CSA	cross sectional area
DLE	dry low emission
DLN	dry low NO_x
EQR	equivalence ratio ($AFR_{stoichiometry}/AFR_{operating}$)
i.d.	internal diameter
IGV	inlet guide vanes
LPP	lean partial pre-mix
NGV	nozzle guide vanes
NO_x	oxides of nitrogen (ppmv corrected to 15% O_2)
O_2	oxygen (%)
OTDF	overall temperature distribution factor
ppmv	parts per million by volume
PRZ	primary reaction zone
RTDF	radial temperature distribution factor
TBC	thermal barrier coating
TCD	compressor delivery temperature
TET	turbine entry temperature
UHC	unburnt hydrocarbons (ppmv corrected to 15% O_2)
VGV	variable guide vanes

1 INTRODUCTION

Over the past 50 years, development in gas turbine technology has changed step by step towards less pollution and more reliable risk-free components. Nonetheless, combustion engineers still continuously face design challenges for higher efficiency, better energy management and conservation, especially with the outcry about gradual deterioration in the air quality and the discovery of an ozone layer hole. Advancements in the conceptual technological approach are still a red alert priority for engineers and scientists in order to meet rising demands for stringent current and projected emissions regulations. ABB ALSTOM POWER UK Limited met the challenges in the environmental aspects by forming an advanced technological team of multidiscipline engineering skills to concurrently work together to spearhead the tasks. The team has not relaxed the more conventional requirements, such as reliable ignition, run-up efficiency, stability and pattern factor, but has concentrated on the wider range of a turn-down ratio implied by current and future legislation to achieve better combustion efficiency at part-load conditions. Consequently, the combustion chamber size and/or better aerodynamic structures inside the combustion chamber may be dictated by pollution constraints.

It was also a design brief that higher turbine temperatures than previously experienced would be employed to improve performance. Such an advancement has to include customer expectations with regard to improvement in reliability, durability and fuel flexibility, and a lower purchase cost should not be compromised.

Cannon *et al.* [**1**] proposed a general design specification for industrial dry low NO_x (DLN) that should incorporate the following features:

(a) machine retrofittable,
(b) adaptable to the full industrial range,
(c) maximize design simplicity (for cost, enhanced life

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and reliability) consistent with achievement of mission objectives,

(d) have a development potential for lower NO_x and increased firing temperature.

The Company combustion team broadly concluded from past experiences that the main objective should be to adopt the process of a dry lean premixed combustion system in line with the majority of industrial gas turbine manufacturers. Also, a system with no mechanical moving parts should be employed to control air and/or fuel and achieve the required reliability and cost consideration. The stabilizing mechanism should be one that imposes a rapid, intense mixing regime with high intensity of heat release in order for the generated NO_x and CO to be controlled in the shortest possible residence time.

It is a known fact that NO_x emission can be reduced by using lean well-mixed primary zones, but the level of reduction is closely related to the quality of the improved fuel and air mixing [**2**, **3**]. Low emissions combustion systems have been demonstrated using radial swirler passages as a means to partially premixing fuel and air for both gaseous and liquid fuels [**4**–**6**]. In a previously investigated radial inflow swirler [**7**], the radial passages have been carried through a bend into an annulus around a central fuel injector, thus becoming a form of axial swirler but with the benefit of a greater passage and mixing time. The complexity of the system provides a very hard task to stop any premature ignition due to a locally high residence time zone that could lead to ignition.

Conventional combustor designs have a very poor mixing in the primary zone to give a wide stability range. This results in very large temperature gradients that generate large NO_x concentrations, as shown in the traverses of Heitor and Whitelaw [**8**]. Future engines, with higher turbine entry temperatures, will require more air in the primary zone and better fuel–air mixing. The concept of utilizing the passages of radial inflow swirler for partial mixing offered low NO_x emissions without using fully premixed fuel and air. Previous work demonstrating an ultra-low NO_x primary zone for gas turbine combustor applications was investigated by Alkabie [**6**].

Alkabie and Andrews [**4**, **5**, **9**–**13**] have examined three different modes of fuel injection: central radial outwards, radial swirler passage and peripheral radial inward injections. The fuel injection in the vane passages of radial swirlers improved the fuel and air mixing compared with the central injection. The internal gas composition measurements showed that for the passage fuel injection the fuel and air mixing was good quite close to the swirler, i.e. a low NO_x level was achieved in a short residence time, and that this was the key to the low NO_x emissions. Also, the swirling jet in the extended prechamber of the swirler exploited the centrifugal mixing effects due to the density differences between burnt (vortex central core region) and unburnt gases to enhance the mixing even further. The results showed that passage fuel mixing could give the required NO_x emission with a reasonable stability margin and it was better than the peripheral or wall injection [**4**, **12**].

The present dry low emission (DLE) radial swirler utilizes the benefit of the passage fuel injection method, which offers enough residence time for the initial partial fuel–air premixing in the passages before introducing the mixture into the prechamber. The geometrical size of the prechamber (e.g. length, diameter) may affect the swirling flow vortex breakdown.

The investigation carried out by Andrews, Alkabie and co-workers [**4**–**6**, **9**–**16**] demonstrated that with more fuel holes located in the upstream region in the passages of the swirler and other stabilizing mechanisms, an improvement in the mixing process could be achieved. In this way more fuel–air uniformity could be achieved to eliminate any local rich zone that may cause a rise in local reaction temperature and hence more thermal NO_x emission via the Zeldovich mechanism. This is in total agreement to what Miknus, Appeleton, Heywood and Hicks [**17**, **18**] have indicated. Their investigation suggested that as the mixing parameter(s) approaches the zero value the peaks of the fuel–air distribution approach the mathematical means of the mixture and low NO_x emission is achieved.

In previous investigations [**10**, **12**], a prechamber length of 40 mm, having the same diameter as the swirler exit, was enough to introduce more mixing length and to delay the breakdown of potential swirling core mass flow caused by the expansion and imposed pressure redistribution. The NO_x level exhibited was in the range of 1–3 ppmv (corrected to 15% O_2) using the atmospheric rig facility [**4**, **10**]. The present work adopted a similar approach to achieve rapid mixing and stability. Furthermore, the choice of the entry and exit profiles of such an extended swirling flow piece proved to be the critical parameter in the aerodynamic stability of the system. The combustion system can only react to what the upstream regions impose on it. The flame anchoring mechanism can be disrupted by the imbalance between the rate of removal of products from the recirculation zone and the rate of burning [**19**].

2 DESIGN METHODS

The geometry of the stabilizing mechanism can influence the combustion characteristics of the system. The entry leading edge blockage and flow separation occurring in some geometry influences the pressure drop and hence the intensity of the combustion zone turbulence. The scale of the turbulence may also be influenced by other integral parts of the flame holder. The residence time within the recirculation region can influence the temperature and mixture ratio limits beyond which combustion cannot be sustained. In lean combustion systems

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where a significant proportion of air is entrained within their circulation region, the residence time can have some measurable effect on the rate of NO_x and CO emissions. Therefore, the flame stabilizer design must satisfy the following basic requirements:

1. The stabilizer drag coefficient and total blockage including the pressure drop due to heat addition must not exceed an acceptable level. Higher values may lead to loss in overall engine performance.
2. The flow regime imposed by the stabilizer must provide ignition interconnection where multiple independent areas of flow recirculation are generated, i.e. to allow flame propagation from one recirculation zone to another during the initial light-up and run-up to load condition.
3. The velocity and blockage created by the stabilizer and any of its integral parts must result in a combustor entrance velocity that is sufficient to preclude flashback under all operating load conditions.
4. The hot product residence time resulting from the aerodynamic structure imposed by the stabilizer should be adequate to produce a steady and stable source of high-temperature gas and active chemical species under all operating conditions.
5. The stbailizer must be self cooling when heat is transferred to it by the burning gases due to radiation, convection or conduction.
6. The stabilizer must permit future expansion without major impact on other parts of the combustion chamber and engine casing.

There are other requirements related to the hot parts life and the acceptable pattern factor for the nozzle guide vanes (NGV). However, it is self-evident that the success or failure of a gas turbine is determined by careful and skilful engineering of the combustion can and the transition piece. The combustion system must deliver a desirable temperature distribution at the turbine inlet. The turbine nozzle is the hottest part of the gas turbine. Its life is limited by thermal fatigue and oxidation; therefore, temperature gradations must be minimized in both the circumferential and radial directions. There are other elements that contribute to the temperature profile such as fuel quality, engine air or fuel maldistribution, i.e. can-to-can variation and air leakage.

The DLE combustion system design work and progress was born by examining other workers' past experiences and published works. In addition, the Company utilized in-house technology advancement and the pre-commissioning development work to satisfy the above criteria. The current DLE main design feature and principal combustion approach may be summarized as follows:

(a) lean burn,
(b) radial swirler stabilized flow with pilot assistance for turn-down,
(c) programmable main/pilot splitter valve in the control system with minimum pilot fuel feed at full-load operating condition,
(d) variable mixing with passage fuel injection regime,
(e) retrofittable replacement for the conventional diffusion flame combustor,
(f) impingement cooled with effusion cooling on selective parts.

The Company's main combustion design parameters were always evaluated before any design release, either experimentally or mathematically, through one-dimensional analyses or whenever practically possible through CFD work. The quality of the assessment of these parts is represented by the level of aerothermodynamics performance of the combustion system on the development bed and field trial. The Typhoon DLN combustion system, as schematically illustrated in Figs 1a and 2, encompasses six reverse flow tubular combustors connected by means of fixture upstream and downstream sections by the effusion-cooled transition duct to the centre casing. A lean premix main burner is employed to reduce flame temperature and hence suppress NO_x production. Furthermore, for ignition and flame stability, particularly during load rejection, a central pilot injector is fully integrated and utilized. This gives full-coverage, double-skin impingement cooling combined with sprayed thermal barrier coating. The thermal barrier coating (TBC) layer acts to lower the inner liner wall temperature to an appropriate level. The spent air travels in the streamwise direction along the annulus, which subsequently vents through a set of dilution holes to mix with the main stream of hot gases. The Typhoon engine combustor is designed to operate with a slightly higher pressure loss to improve mixing and cooling effectiveness of the impingement liner. A higher percentage air flow is passed through the swirler, giving a lower primary reaction zone equivalence ratio of $0.5 \leqslant \phi_{PRZ} \leqslant 0.6$.

For gaseous fuel operating at the above range, a reaction temperature $\leqslant 1800$ K is needed to achieve a low thermal NO_x production. A narrow temperature band does exist where NO_x and CO emissions may be minimized for premix lean burn systems. However, there are some limitations on the extent to which the flame temperature can be lowered. Operating near the limit of the lean blowout can cause flame instabilities and a rapid decrease in the CO depletion rate without the assistance of a highly reliable pilot burner.

The Tornado combustion system as shown in Fig. 1b consists of eight reverse-flow tubular combustors. The burners used are identical to those employed by the Typhoon. The combustion chambers, however, are different. For reasons of space required for the burners, the Tornado combustion chamber is the same diameter and about half a diameter longer than those for the Typhoon. To accommodate the retrofit DLN hardware, a

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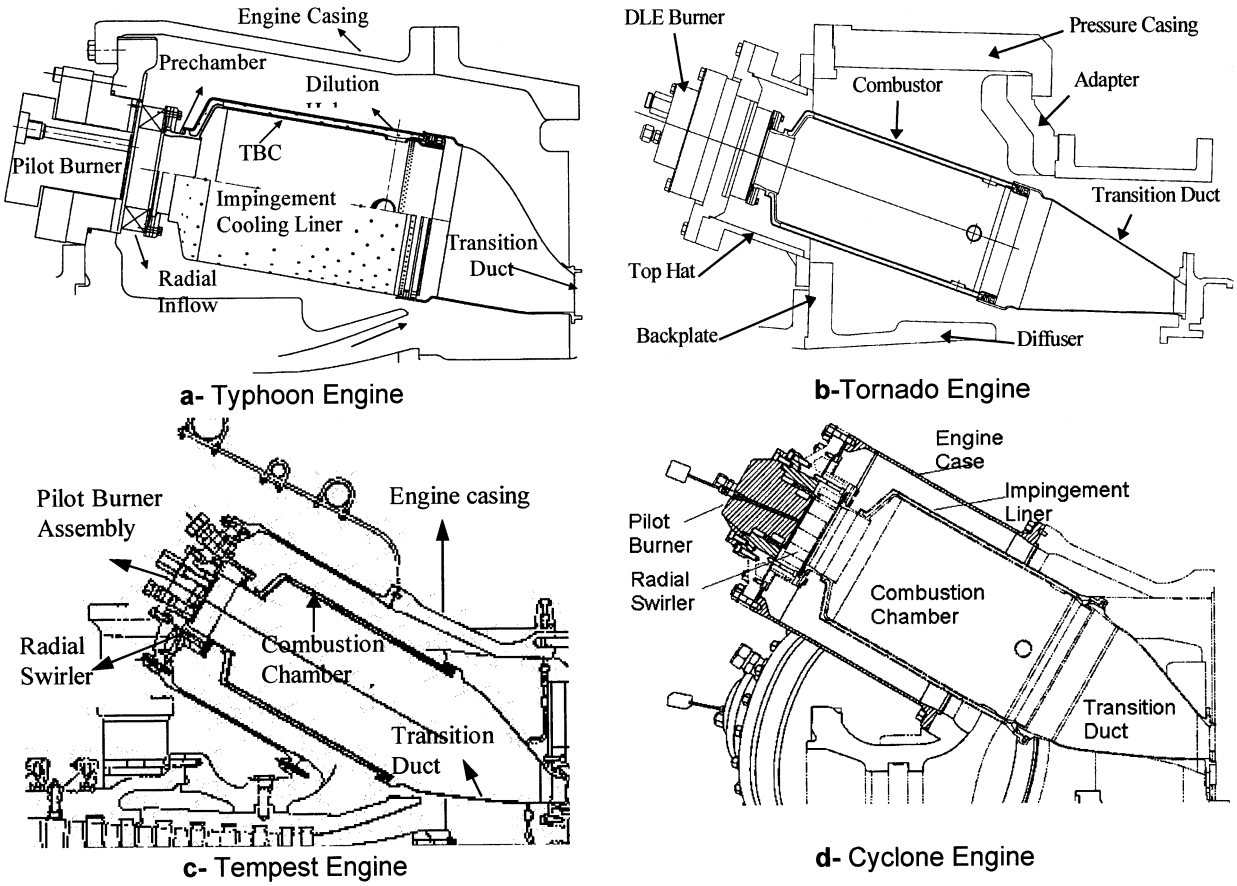

Fig. 1 Schematic of the DLE combustion system layout for various engine ratings

new centre case was designed. The lower compressor discharge temperature and pressure for the Tornado compared to the Typhoon, along with the higher air mass flow per combustor, led to the primary reaction zone equivalence ratio ϕ_{PRZ} being increased by 11 per cent to that of the Typhoon primary zone region to achieve the design turbine entry temperature (TET). However, this was simply achieved by an increase in the transition duct effective area [**20**].

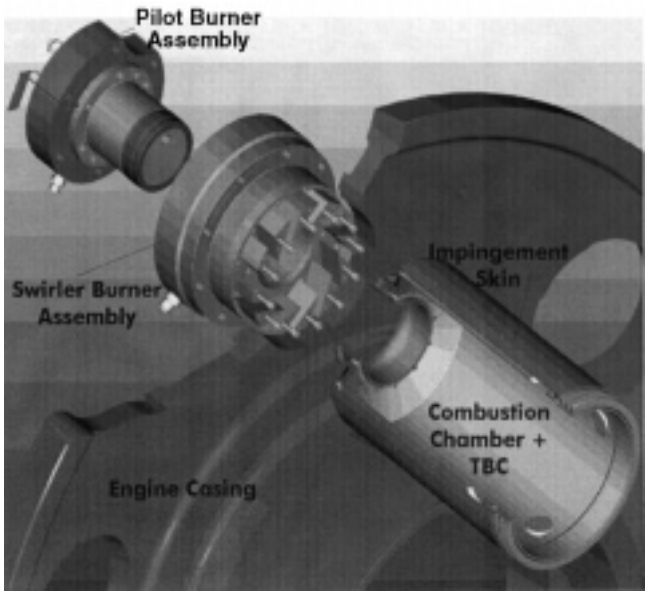

Fig. 2 Cutaway of the Typhoon combustion system assembly

The Tempest DLN combustion system as illustrated in Fig. 1c encompasses six reverse flow tubular combustors connected by means of a transition duct to the centre casing. The Tempest is a 25 per cent scale-up of the Typhoon and is designed to maintain the Mach number in the prechamber, aspect ratio of the swirler slots, length–diameter ratio and expansion ratio of the combustor to minimize any aerothermal differences. As with the Typhoon, a lean premix main burner equipped with a pilot system is employed to reduce flame temperature and hence suppress NO_x production, and also to cover a wide ignition and flame stability margin, especially during low-load conditions.

Figure 1d shows the schematic of another standard DLE combustion system employed to power the Cyclone engine. The Cyclone has six reverse-flow tubular combustion chambers located around the high-pressure casing. The combustor can operate on natural gas or distillate-2 fuel. The programmable control system provides all necessary control and protection for smooth

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changeover between primary and secondary fuel at any load. The Cyclone engine is a 13.4 MW (18 000 hp) type industrial gas turbine suitable for electrical power generation and mechanical drive. The engine has a simple cycle efficiency exceeding 35 per cent. The engine offers cost effective power for a multitude of duties, including offshore platforms, industrial power generation and pipeline transmission.

3 COMBUSTION SYSTEM TEST LOGIC

Figure 3 illustrates the ABB ALSTOM POWER logic flow diagram for testing the combustion system. The full-scale combustor tests are used to verify the ignition, run-up efficiency, emissions and heat transfer analysis at realistic simulated and/or engine operating conditions. Geometrical calibration is under continuous assessment to ensure that the design intent has been met. By far the most important aspect in the DLE combustion system development work is engine air management. Air percentage and placement will almost guarantee a healthy combustion process and engine performance. It becomes common practice for the company combustion team to identify the air mapping for each of the DLE configurations.

The process starts by identifying how much air is passed by the compressor discharge shroud. This value may be derived from a performance programme that identifies the overall engine specification. However, the combustion engineer is constrained by the latter design criteria requirements in order to achieve a certain commercial target in the market place.

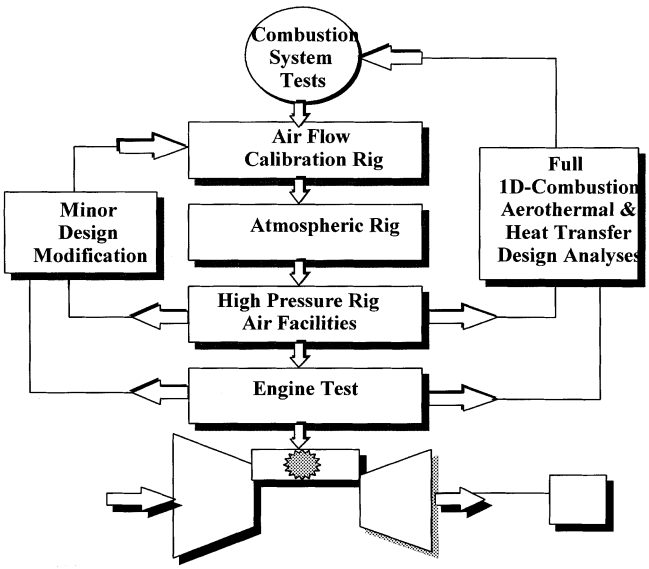

Fig. 3 Schematic simplification of ABB ALSTOM POWER combustion system test logic

From the one-dimensional analyses, the operating swirler and primary zone Mach number can be identified for the first iteration and hence the combustor diameter and length calculated. A typical industrial reference Mach number for the primary zone region should not be >0.047, based on the combustor diameter. With a given fuel specification, the reaction temperature can be calculated together with the combustion velocities and other aerothermodynamic parameters can be identified. All other procedures are iterative that may be modified and fine tuned using atmospheric and high-pressure air facility rig results.

4 COMBUSTION SYSTEM AIR MAPPING

It is paramount to understand the details of the engineering design of each of the components that is encompassed by the combustion system. Then engineering efforts should be directed towards calculating each individual item's effective open area and its position in relation to the incoming flow. One-dimensional analysis may guarantee the evaluation of the orifice discharge coefficient to within ± 2 per cent. However, a more robust approach is to validate each geometrical result discharge coefficient by placing the actual configuration in an automated isothermal air flow rig and calculating the average mean of each item. The following needs to be considered:

$$A_{\text{geometry effective area}} = \sum_{i}^{n} A_{e_i}$$

where $i = 1,2,\ldots, n$th component that is encompassed by the combustion system. Then

$$\sum_{i}^{n} \frac{A_{e_i}}{A_{\text{geometry effective area}}} + A_{\text{leakages}} = 100\%$$

The air leakage is based on the best design available in the market. The severity of the uncontrollable air leakage is of primary importance. Too much leakage from the downstream section of the burner, or even before, could cause serious problems in terms of component life, stability and emission penalty, especially that of the NO_x level. Any reduction in the amount of design air requirement can jeopardize the stoichiometry of the primary reaction zone and hence produce a high reaction temperature. Therefore, an engine engineering design assembly stack-up and visual check is the best way of reducing the impact of undesirable leakage.

Ideally, each individual component will shear an amount of air that is directly proportional to the ratio of the effective area of the combustion system geometrical

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total. Therefore,

$$M_{\text{system total}} = \sum_{i}^{n} \left(\frac{A_{e_i}}{A_{\text{geometry effective area}}} \right) m_{\text{total air}}$$

where $M_{\text{system total}}$ is the total mass flowrate delivered to the combustion system and A_{e_i} is the individual component effective area.

5 PRESSURE DROP AND MACH NUMBER

The flow field through a radial inflow swirler has a very complicated structure and is dependent on the geometrical shape of the swirler [**6**]. The flow suffers separation at the swirler entry which causes a reduction in the open flow area. This is why such a type of aerodynamic stabilizer has a low discharge coefficient. The level of turbulence imposed on the combustion domain controls a large part of the rate of fuel–air mixing and the subsequent rate of flame propagation. However, the level of turbulence is controlled by the combustion system's burner pressure drop and blockage.

The primary zone Mach number is a known factor influencing the residence time and hence the emission concentration. It can be employed as a combustion chamber scaling parameter. The relationship between the total overall pressure loss and primary zone Mach number is expressed as follows [**6**]:

$$\frac{\Delta P}{P} = \frac{\gamma}{2} \left[\left(\frac{M_{1_{PRZ}}}{C_{D_{\text{geometry}}}} \right) \left(\frac{A_{\text{combustor}}}{A_{\text{swirler}}} \right) \right]^2$$

where $\gamma = C_p/C_v$, $M_{1_{PRZ}}$ is the approach Mach number and $C_{D_{\text{geometry}}}$ is the geometric discharge coefficient. The level of turbulence generated by the mass of air flow through the swirler is controlled by the burner pressure loss, and the velocity gradient in the jet shear layers increases as the burner blockage increases [**9**]. Hence, the fluid dynamics of the shear layers and subsequent level of turbulence generated are interlinked.

The work and investigation carried out by Alkabie and Andrews [**9**, **10**] on the radial inflow swirler suggested that in order to achieve high combustion efficiency the overall pressure loss has to be $\geqslant 4$ per cent. The combustion efficiency exhibited was better than 99.99 per cent. However, Swithenbank [**21**] suggested that the minimum condition for a perfectly stirred reactor may be established at $\Delta P/P_t \geqslant 3.0$ per cent. The energy balance shows that the pressure loss factor is made up of the sum of losses associated with distortion of the mean velocity profile, the generation of turbulence and the total dissipation of the turbulence energy. The total pressure loss may be written in the following format:

$$\frac{\Delta P_{\text{combustor}}}{P_t} = \frac{R}{2} \left(m_{\text{air}} \frac{\sqrt{T_2}}{P_2 A_e} \right)^2 + \left(\frac{\Delta P}{P_t} \right)_{\text{heat addition loss}}$$

The pressure loss due to heat addition or Rayleigh pressure loss is a function of Mach number within the combustion chamber and temperature rise [**22**]. However, the Mach number approaching the flame front is typically less than 0.05. Thus, heat addition pressure losses are generally very low compared with diffuser and liner pressure drops, although it is desirable to minimize $\Delta P_{\text{diffuser}}$ since this is a parasitic loss. Nonetheless, the $\Delta P_{\text{combustor}}$ is used to promote uniform air distribution and provide high swirler and dilution air velocities for uniform and intense mixing within the combustor. It may be simplified and expressed as follows [**22**]:

$$\left(\frac{\Delta P}{P_t} \right)_{\text{heat addition loss}} = \left(\frac{T_2}{T_1} - 1 \right) \left\{ \frac{\gamma M_1^2}{2} \left[1 + \left(\frac{\gamma - 1}{2} \right) M_1^2 \right]^{\gamma/(\gamma-1)} \right\}^2$$

6 SWIRL NUMBER

In order to characterize the Company DLE radial swirler, a derived geometrical swirl number S_g [**4**,**6**] is employed to evaluate the strength of the rotational flow regime provided by the stabilizer. It can be used as a design tool to calculate specific designs of the radial swirler and combustion chamber. The swirl number at the prechamber can be evaluated using the following equation [**6**]:

$$S_{\text{prechamber}} = \left(\frac{\sin\theta}{1 + (\tan\theta)^{-1}} \frac{A_p}{A_{e\ \text{swirler}}} \right)$$

Furthermore, the Swirl number at the combustor entry is satisfied by the following correlation, which includes the effect of the vortex breakdown immediately after the expansion:

$$S_{\text{combustor}} = S_{\text{prechamber}} (d/D)$$

where

θ = swirl angle (rad)
A_p = prechamber cross-sectional area (CSA) (mm^2)
$A_{e\ \text{swirler}}$ = swirler effective flow area (mm^2)
d = prechamber i.d. (mm)
D = combustor i.d. (mm)

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7 COMBUSTOR REFERENCE AND MAXIMUM VELOCITY

Two velocities characterize the flashback or burnback conditions. The reference velocity V_{ref} is defined as the combustor mass flow divided by the inlet air density and maximum combustor cross-sectional area. The maximum axial velocity V_{max} is defined as the maximum value of the flow axial component of velocity at the prechamber exit or combustor inlet.

In general, the reduction in the combustor reference velocity is found to produce two modes of failure in the premixed combustion system, depending upon the stabilizer geometry. In the flashback condition, the flame is observed to jump sharply upstream, attaching itself to the fuel injection tubes in the premixed section. In the burnback situation, at low reference velocity the flame stabilizer and any attached parts apparently may melt from the inside, causing mechanical failure. The low reference velocity could result in an increase in the regional temperature and a decrease in the external convective cooling. The two velocities are related by the following equation:

$$V_{max} = f(\Delta P_{overall}) = \frac{V_{ref}}{1 - B/100}$$

where B is the blockage to the inlet air flow by the stabilizer. The ability to withstand flashback appears to be a weak function of the flameholder geometry. The flashback velocity may be represented linearly by the following equation [**23**]:

$$V_{flashback} = C(\phi_{operating} - \phi_{WE})$$

where C is a constant $= 123$ and ϕ_{WE} is the equivalence ratio at weak extinction (for fully premixed ~ 0.41).

The above correlation is a result of a premixed stream burnt at an inlet pressure of 0.56 Pa and inlet temperatures of 610 and 700 K in a sudden expansion (dump) combustor, and of measurements made of flashback velocity as a function of the gas stream equivalence ratio. In an effort to achieve a robust DLE design, V_{max} is always kept $\gg V_{flashback}$ across the power range for both gaseous and liquid fuel operations. The DLE system depends upon the flow aerodynamics structure, air and fuel placement and a local pressure drop to reduce or eliminate the harmful pressure fluctuations. The latter may happen as a consequence of vortex breakdown. The precessing vortex core seems to be associated with vortex breakdown and other oscillations occurring either independently or as consequence of the reverse vortex core [**24–26**]. The energy loading of the intermediate region between two recirculation zones could generate large turbulent shear fluctuation levels and directional intermittence that may largely contribute to random fluctuations and instability [**24**].

8 VARIABLE MIXING CONCEPT

The rapid and intense mixing process is a predominant necessity to accomplish fully premixed fuel and air in a short combustor. This would insure low NO_x emissions and possibly low levels of CO. The company approach to the DLE mixing process is entirely different from the former conventional method. The new approach utilizes the swirler, prechamber and the jet shear layer (see Fig. 4a) in three consecutive mixing regimes, namely partial, premixed and fully premixed. The steps imposed by the new mixing regime may be described as follows:

1. The partial premixed regime utilizes the radial swirler passages where at least 60 per cent of the mixing could be achieved.
2. The premixed regime takes place in the prechamber mixing length to achieve between 15 and 20 per cent of

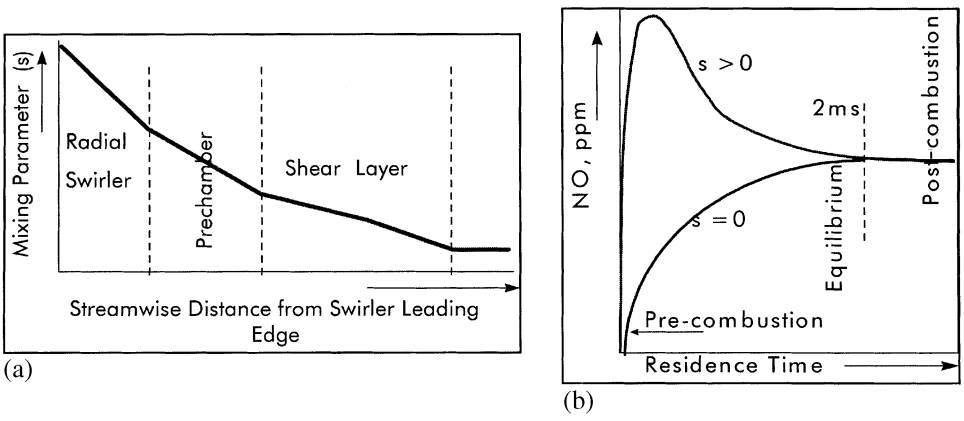

Fig. 4 Schematic representations of (a) the variable mixing concept and (b) the NO generation hypothetical mixing parameter time

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the mixing process due to the rotational force imposed by the flow solid body rotational force.

3. The fully premixed mixture is due to the high energy dissipation caused by the vortex breakdown in the combustion chamber. The expansion ratio plays a very crucial part in the final mixing process and heat release to the combustion chamber.

The maximum reaction temperature exhibited by the system is in the centre reverse core region, where maximum thermal efficiency is established. The centre columnar flow is fed by the element breaking away from the high intensity shear layer where maximum heat release may occur. The latter may act as an ignition source for the freshly charged mixture issued by both the swirler and the pilot burner.

The pilot system acts as a source generating a local rich zone that approaches the full characteristics of the diffusion flame. This will ensure flame stability but there is a trade-off in NO_x emission. Figure 4b is a schematic illustration of the hypothesis of the correlation between mixing strength, residence time in the precombustion and NO generation. There is only 2 ms to establish full premixing between air and fuel, i.e. the mixing parameter $S = 0$, and the NO approaches that of the equilibrium level. However, if during that time the mixing parameter $S > 0$ then the system will overshoot, with a spike of NO before it levels off as it approaches the equilibrium value. In a DLE combustion system the effect of the post-combustion condition should not influence NO_x formation. Furthermore, the NO_x level is independent of the combustor overall residence time.

One approach targeted by DLE combustor design is an examination of the non-reacting single and multiple fuel jet mixing in the swirler passages with reference to the air slots cross-flow. Indeed, numerical and experimental studies have demonstrated that an optimum momentum flux ratio does exist for various radial swirler slot geometries. The balance between NO_x and CO production may be attributed to the extent that the jet penetrates compared with the jet boundary issued by the radial slot.

The relation between the maximum penetration Y_{max} exhibited by the fuel jet and the air jet mass flow rate in the slot may be expressed as follows:

$$Y_{max} = CC_f \sqrt{\frac{\rho_f V_f^2}{\rho_a V_a^2}}$$

where C is a constant ranging between $1.05\ C_f \leqslant C \leqslant 1.25C_f$, $C_f = f$ (fuel property and injector geometry) and $\rho_f V_f^2/(\rho_a V_a^2)$ is the fuel–air jet momentum flux ratio. Figure 5 shows an early one-dimensional analysis result for a single jet fuel hole maximum penetration using the above correlation.

A maximization of the jet transfer surface with the cross-passage flow is essential to optimize the mixing regime. A higher momentum flux ratio may cause overpenetration along with the risk of jet backflow. On the other hand, lower momentum flux results in fuel jet

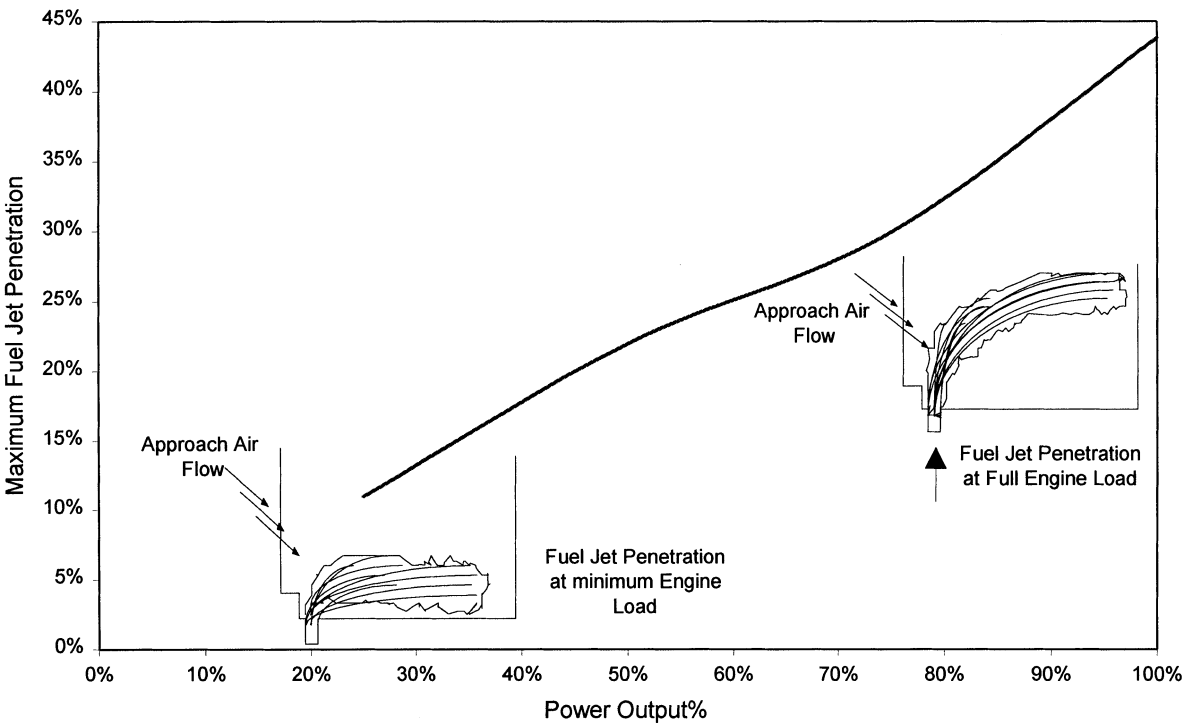

Fig. 5 Early one-dimensional analysis assessment work for maximum fuel jet penetration as a function of the Typhoon engine power output

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underpenetration. The latter may lead to the generation of a localized rich area at the back face of the swirler. This generates more thermal NO production. A fuel jet in a cross-flow, such as a passage injector, may impose a Gaussian distribution about the mean equivalence ratio $\overline{\phi}$ or the mean fuel mass fraction, which are assumed to be almost equivalent; i.e. the fraction of flow with an equivalence ratio or fuel fraction between ϕ and $\phi + \mathrm{d}\phi$ is given by $f(\phi)\,\mathrm{d}\phi$:

$$f(\phi) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right) \exp\left[\frac{-(\phi - \overline{\phi})^2}{2\sigma^2}\right]$$

where σ is the standard deviation.

The mixing parameter s, defined as $s = \sigma/\overline{\phi}$, is employed for the present work as a measure of the non-uniformity of mixing in the primary zone region, where $s = 0$ corresponds to perfect mixing. The mean primary reaction zone equivalence ratio $\overline{\phi}_{\mathrm{PRZ}}$ is determined from the fuel and air flows into the primary zone. However, accuracy of the value depends on the measured fuel flowrate and the calculated air mass flowrates through the front end and across the combustion system. Some empirical correlation may satisfy the NO_x emission level for various configurations. There is no universal formula that may satisfy the NO_x production at various engine load conditions. However, each has its own merit at various load conditions if the various working parameters are fully understood.

Figure 6 shows the NO_x concentration as a function of the mixing parameter(s) and primary zone equivalence ratio. The correlation is based on the theoretical and experimental studies carried out by Heywood and co-workers [**17**, **18**] on NO formation in gas turbine engines. The correlation was modified and validated against the in-house NO_x emission results for various engine configurations and internal traverse results of Alkabie and Andrews [**12**]. One of the most successful correlations used in the one-dimensional analyses may be written as follows:

$$\mathrm{NO}_x = c_1 \exp[c_2 \phi_{\mathrm{PZR}}(1 - s) + c_3 s]$$

where c_1, c_2 and c_3 are constants determined experimentally (typical values are 2×10^{-5}, 19.735 and 30 respectively). The mixing parameter has been treated as a floating parameter whose value is determined by matching predictions to measured emissions data. The effect of the pilot fuel mass flowrate has been proved to have a deterioration effect on the mixing regime and hence on NO_x production. The following correlation takes account of the pilot to main fuel split ratio effect at any thermal load conditions:

$$\mathrm{NO}_x = P_2^n \tau_{\mathrm{mean}} f(\text{Pilot } \%) \exp(cT_{\mathrm{flame}})$$

For a good design of a radial swirler $0.13 \leqslant n \leqslant 0.35$ depends on the configuration. These figures are below the most conventionally employed values of 0.5–0.65. Also,

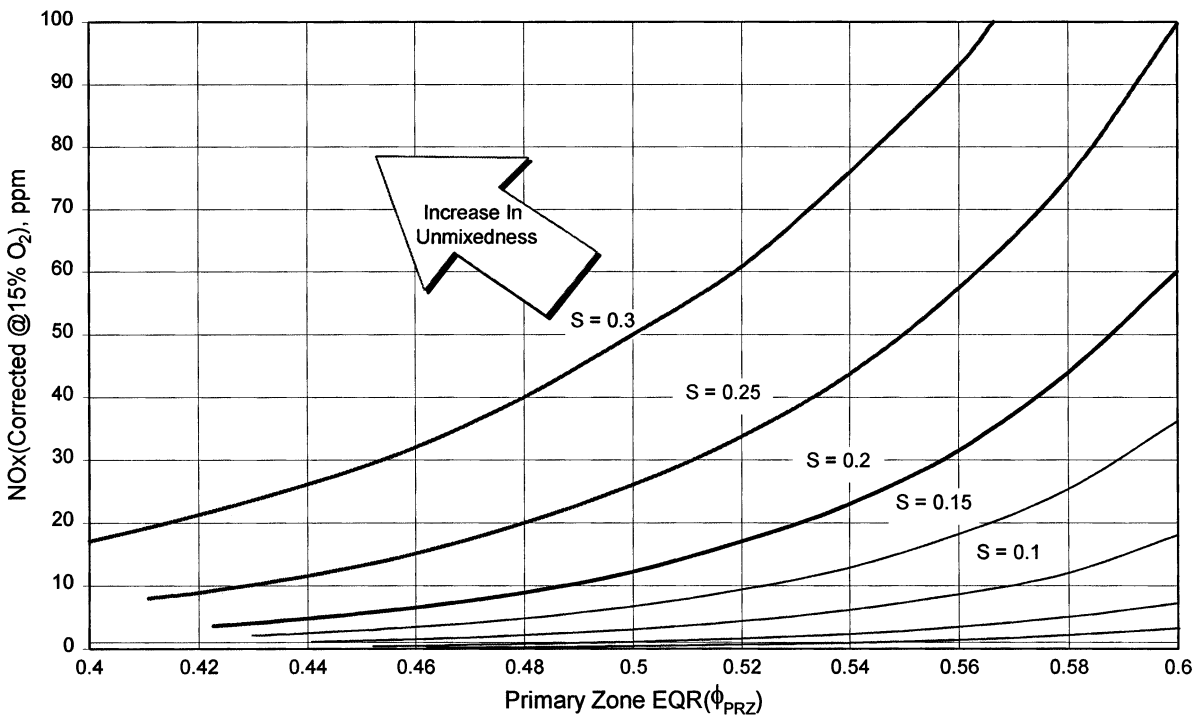

Fig. 6 Mixing parameter effects on NO_x production for various primary zone equivalence ratios

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they were validated against test data from high-pressure rig facilities and later calibrated against engine data from development bed and field tests.

9 NO_x AND CO EMISSION RESULTS

Fuels such as natural gas and gas oil contain no or small amounts of nitrogen. Therefore, NO_x formation in the combustion chamber is caused by a minor contribution from fuel nitrogen, being largely due to the oxidation of nitrogen from the combustion air via the Zeldovich process and the prompt or N_2O intermediate mechanism. The maximum production of NO ends within a residence time of 2 ms for stabilizers such as the radial swirler employed for the present work. The time of 2 ms represents the relaxation time needed for a nitrogen molecule to go from an excited state to an equilibrium one. Moreover, the reaction rate for NO_x formation at postcombustion temperatures must be very slow. This was also demonstrated [**12**] utilizing similar passage fuel injection by traversing a 140 mm i.d. combustor on the atmospheric test rig.

Figure 7 demonstrates typical NO_x and CO emissions for the Tornado (Fig. 7a), Tempest (Fig. 7b), Typhoon (Fig. 7c) and Cyclone (Figs 7e and f) engines as a function of engine power output. Also, shown in Fig. 7d is the emission of the Tornado twin-shaft engine as a function of corrected power output. The length and diameter of the combustion chamber and hence the mean residence time plays little or no part in the NO concentration. However, the deterioration in the mixing regime during part-load operation plus assistance of the pilot fuel produced high values of mixing parameters (see Fig. 6) and hence a rapid increase in the NO_x level.

In a practical engine, it is hard to maintain the same mixing parameter *s* without varying the geometry of the combustion system. There are three conditions that may exist for high CO production:

1. The cool wall of the combustor quenches the hot products that slow or freeze the CO depletion rate.
2. The rapid mixing between hot products with cold gases may be worsened by the shorter residence time.
3. Pockets of unmixed fuel and air escape the hot combustion zone.

For the present work, the first reason may be the dominant factor that contributes to the final CO level at a part-load engine condition. However, the majority could be attributed to the unmixedness and the escape of the unburned pockets from the primary zone region to cause high CO and UHC emissions. There are optimum zone operating conditions for minimizing CO levels across the engine power range, as will be discussed later. For the present configuration, the optimum means that the engine would operate near the full-load design equivalence ratio $\phi_{\mathrm{PZR\ design}}$.

10 EFFECT OF AMBIENT CONDITIONS

Figure 8 shows real-time measurements of Typhoon DLE engine data as being monitored by a continuous emission monitoring (CEM) system from a site in Holland. For this particular site, the emission results are a good demonstration of how little the amblent variations affect the DLE system, even though the fuel properties and atmospheric conditions could have a pronounced effect on the reaction zone temperature and hence the engine performance. The fuel type (i.e. gas or liquid), compressor performance and engine configuration may have dominant effects on the fuel vaporization rate and fuel jet penetration and hence the mixing quality in the primary zone region, as shown in Fig. 9. The variation in the ambient temperature was found to affect the pressure fluctuation, especially that of low frequency in the range of 68 Hz or so, called 'chug' dynamics at the base-load condition. Also, the same effect was reported on-site with Typhoon engines fuelled with a high Wobbe number [**22**]. The Company has found the remedy to the problem by fuel schedule reiteration for each specific site. The harmonization in the overall engine-controlled management, which includes fuel and air split mapping, is crucial to suppress all engine operation side-effects.

11 STABILITY LIMIT AND VGV MODULATION

Sometimes lean stability limits do result in a clear blowout point, but can be characterized by the excessive emissions of CO and UHC species that saturate the analysers before combustion actually ceases. The Company DLE system utilizes a pilot burner to extend the lean limit across the power range. Figure 10 shows a typical example of the integrated action that the pilot has to go through during Typhoon engine changing load conditions. This is carried out so that the local adiabatic flame temperature increases. Therefore, the lean stability limit is primarily a function of the adiabatic flame temperature. The lean premixed combustion system inherited a low lean stability limit. Engine operations are constrained and could be very limited to the band-specific load demand and operability. This may be typically characteristic of the lean partial premix (LPP) due to the slow CO depletion rate. The CO burnout rate is reduced rapidly near the blowout limit. However, with the correct pilot fuel schedule the rate of CO burning can be accelerated, but not indefinitely, to cover the range of part-load

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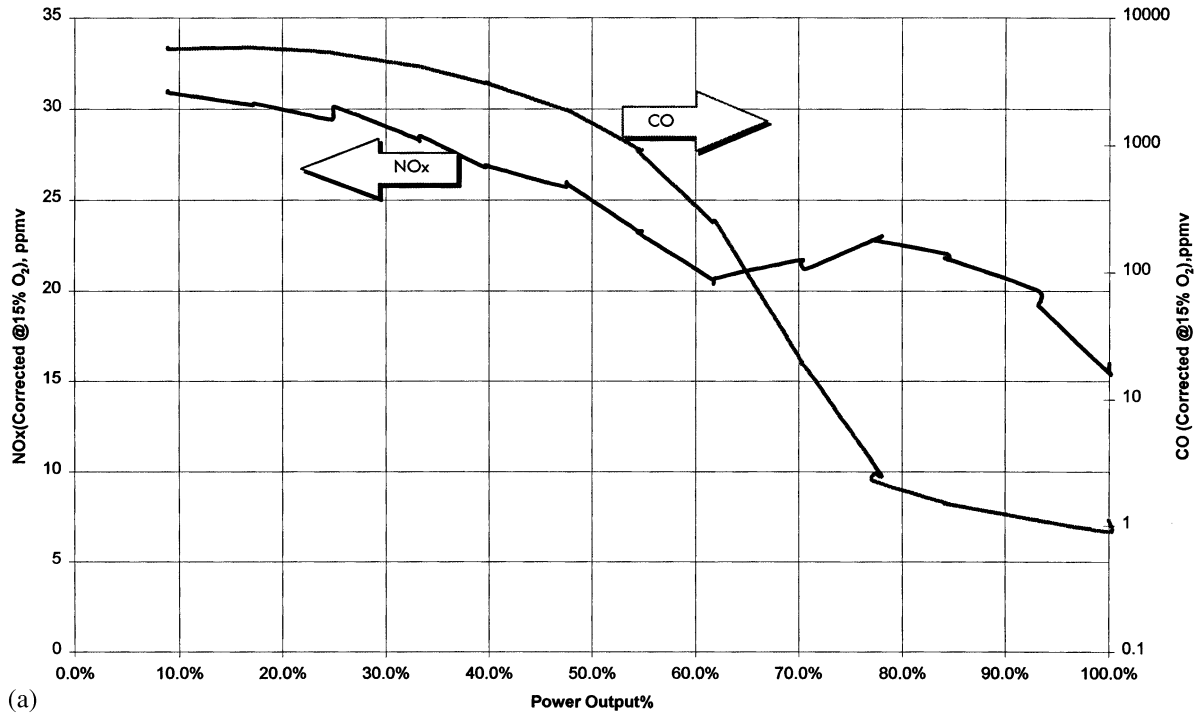
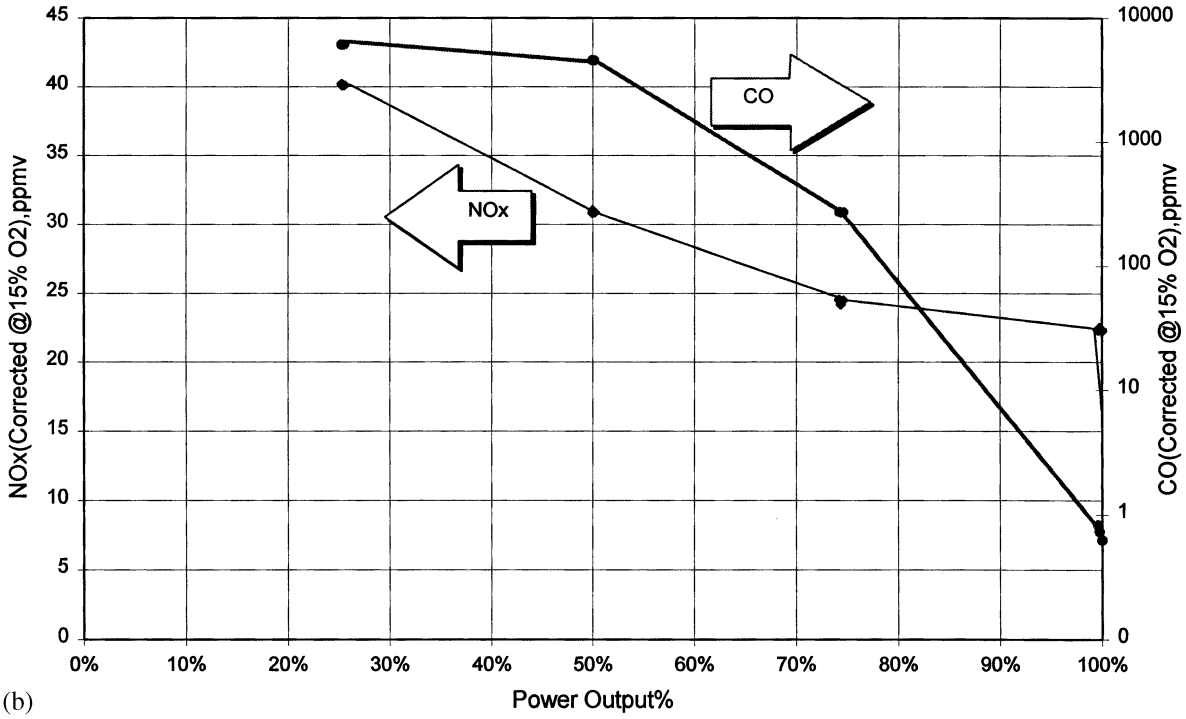


(a)



(b)

Fig. 7 (continued over)

operation due to combustor overloading, therefore causing a rise in the combustion inefficiency.

The present DLE system can deliver a turndown ratio of 10–15 per cent at approximately 85–100 per cent load conditions without the aid of moving mechanical parts. Substantial CO emission reduction can be effected by variable guide vanes (VGV) modulation. The modulation provides rapid combustion, yet quickly brings the equivalence ratio to some intermediate value depending on the configuration and design ϕ_{PZR}. This action inhibits NO formation without slowing the CO oxidation process to frozen conditions. One-dimensional analysis indicates

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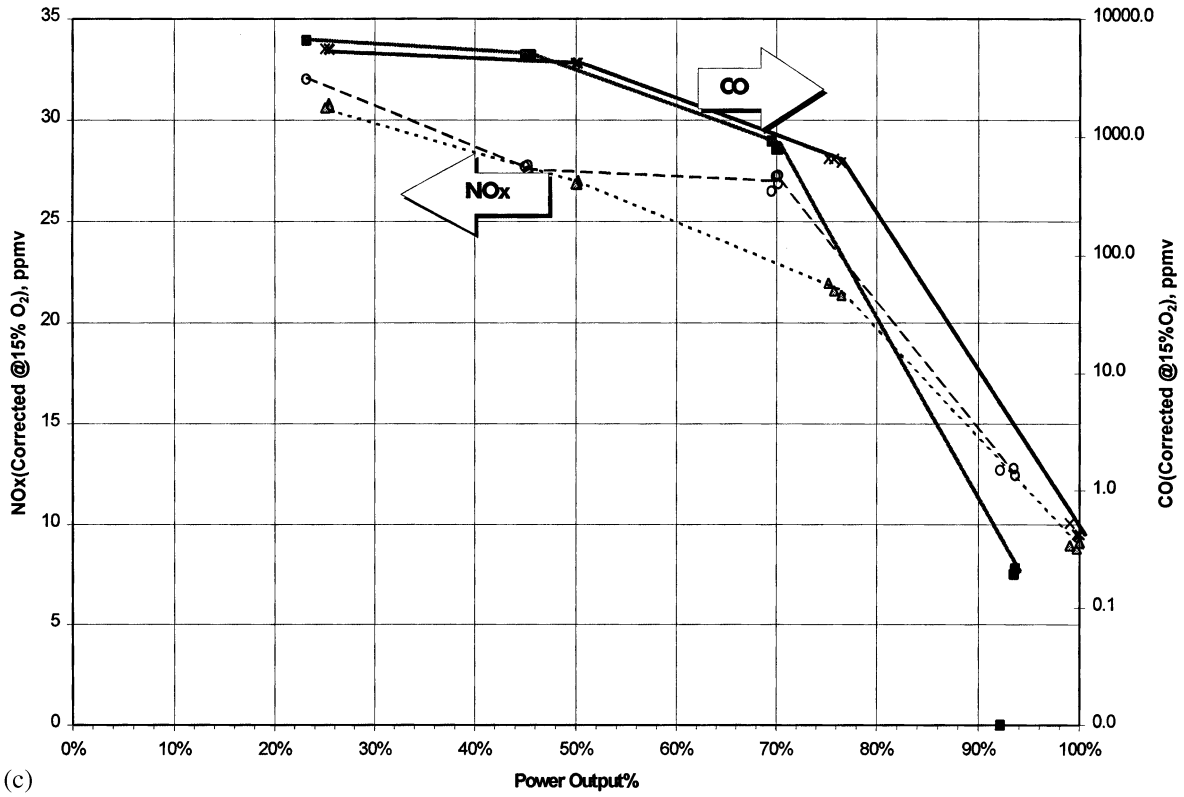
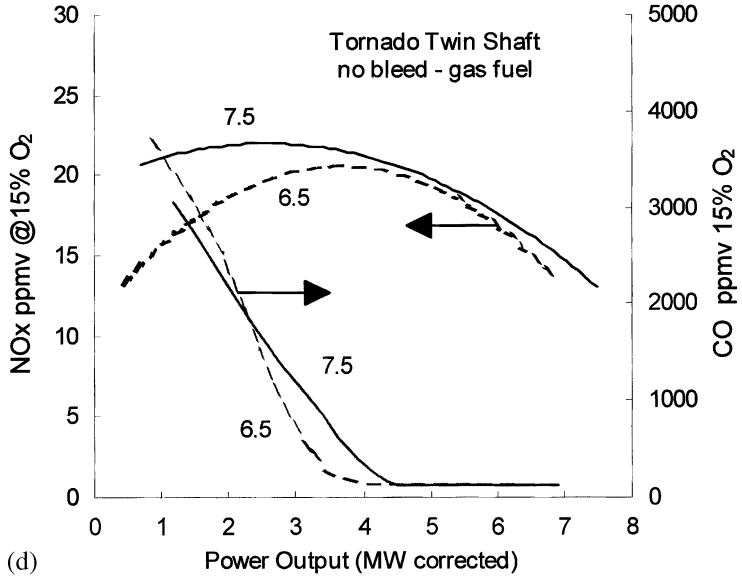


(c)



(d)

Fig. 7 (continued over)

that an intermediate value of ϕ_{PZR} of 0.5, i.e. a gas temperature of ~1700 K, would allow CO conversion to CO_2 without the penalty of increasing NO concentration.

The total air supplied by the compressor can be modulated via the inlet guide vanes (IGV) to assist in maintaining the primary zone stoichiometry as close to the full-load design point as the mechanical design and compressor surge point permit. Figure 11 shows the emission results due to VGV modulation on a dual-fuelled Typhoon engine for a power range between 60 and 85 per cent load conditions using one of the development test beds at the Lincoln site. The results were as expected due to two factors: firstly, the rise in the reaction temperature can accelerate the depletion of the CO level and, secondly, an improvement in the mixing rate can occur as the air–fuel mixture approaches the

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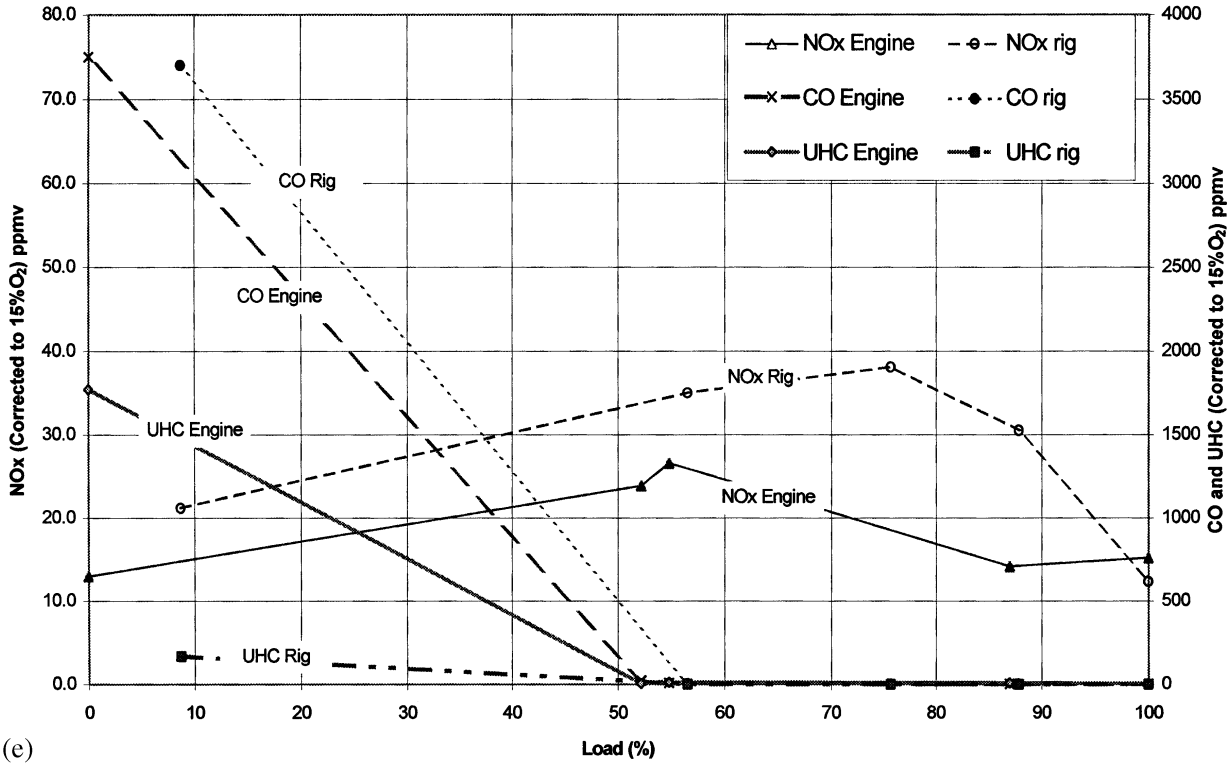
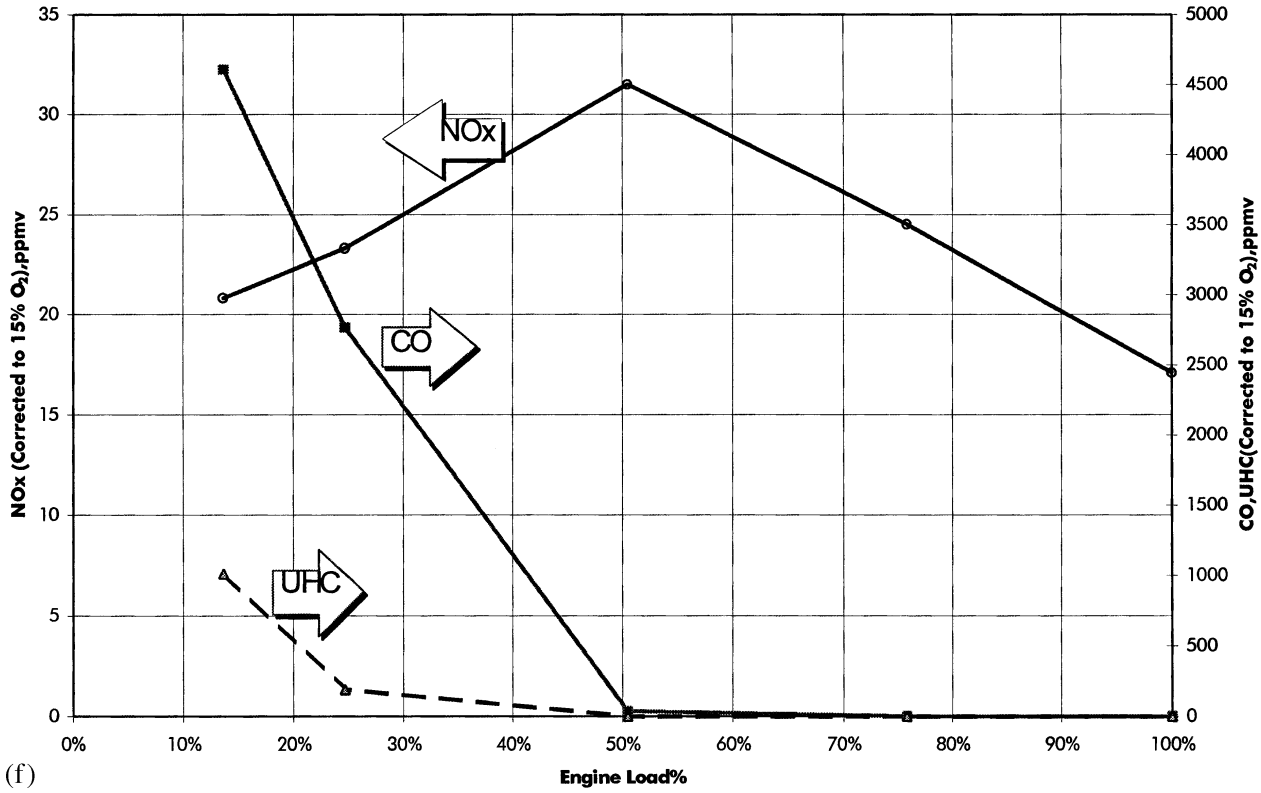


(e)



(f)

Fig. 7 Variations of NO_x and CO emissions with (a) Tornado engine and (b) Tempest engine power output (single-shaft gas). (c) NO_x and CO emissions as a function of Typhoon engine power output (gas); the pilot split is not optimized. (d) NO_x and CO emission variations with Tornado (twin-shaft gas) engine power output. (e) Variations of the NO_x, CO and UHC emissions with the simulated and actual Cyclone production engine power output (gas). (f) NO_x, CO and UHC emissions as a function of Cyclone production engine power output without air bleed (gas); the pilot split is not optimized

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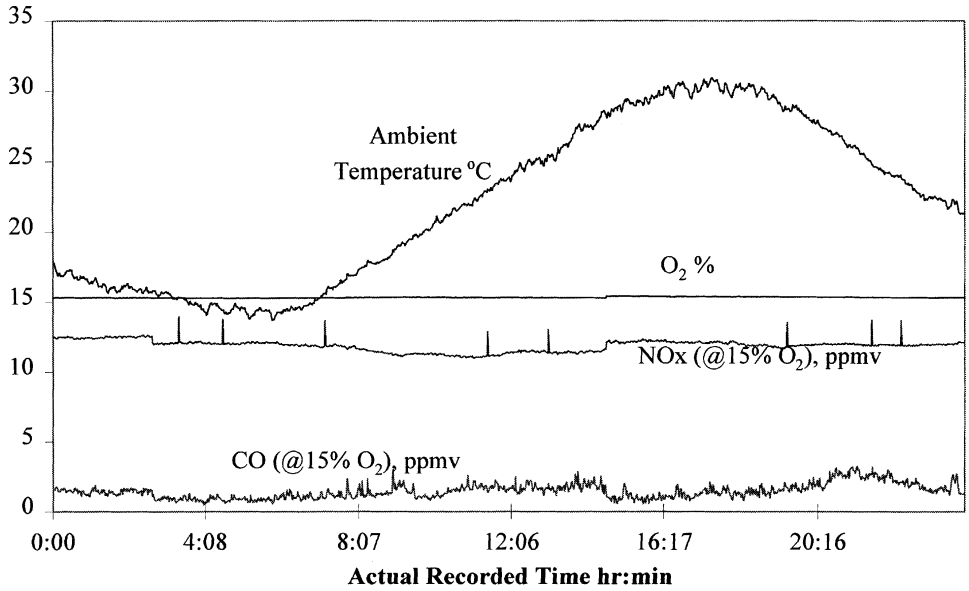

Fig. 8 NO_x and CO emissions as a function of the ambient condition for a single-shaft Typhoon engine

design operating conditions. The variable guide vanes modulation improved the turndown ratio by $\geqslant$40 per cent. In some cases, the compressor characteristic may impose a certain flow field that is not suitable for combustion system performance. Indeed, fast acceleration, deceleration or the transient cyclic effect can rapidly downgrade the combustion efficiency characteristics. Numerical results show that at high Reynolds number and low laminar burning velocity, the combustion process is extinguished.

A superior turndown ratio is demonstrated by the Cyclone engine down-load at 50 per cent and without any casing air bleeds. This is attributed to the high depletion rate of CO due to a high reaction temperature and pilot split, as shown by Figs 7e and f. However, the wall temperature was kept high enough not to cause any reduction or freezing in the reaction zone.

12 COMBUSTION SYSTEM COOLING

The ceramic TBC concept provides a means of raising the allowable operating component temperature while em-

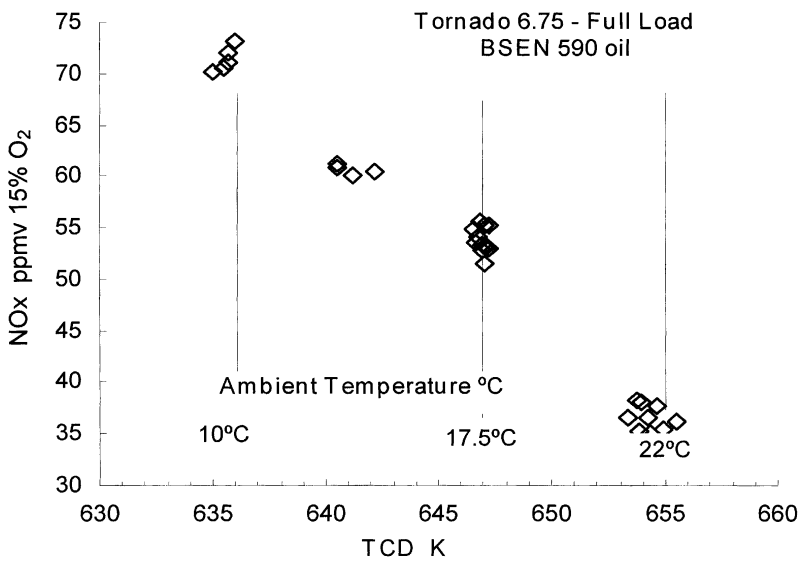

Fig. 9 Variation of NO_x with the compressor discharge temperature shows the ambient condition effect on the single-shaft Tornado engine

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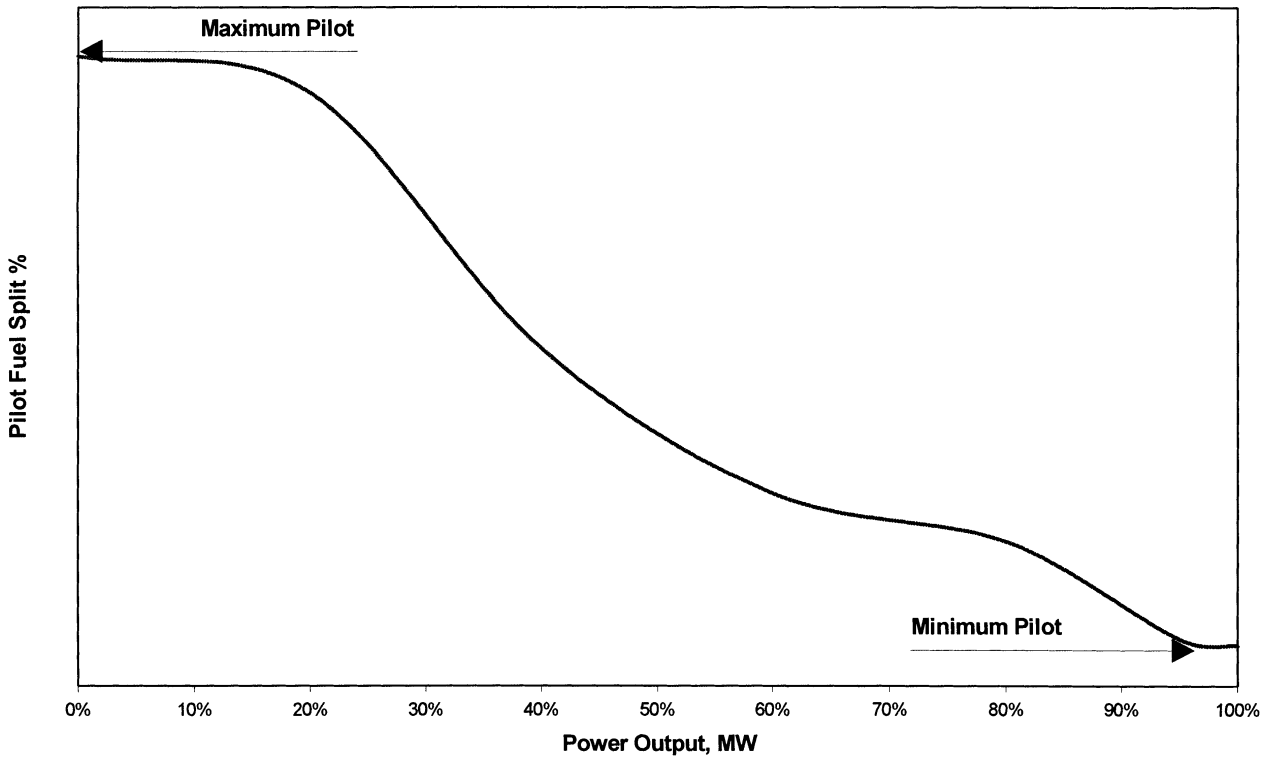

Fig. 10 Typical example of a pilot fuel schedule at various load conditions

ploying the currently available metallic materials. The TBC should satisfy several requirements including a high melting point, a low thermal conductivity, a flame reflectance higher than the substrate material and a thermal expansion coefficient similar to the substrate, and without interfering with the component aerodynamics. In

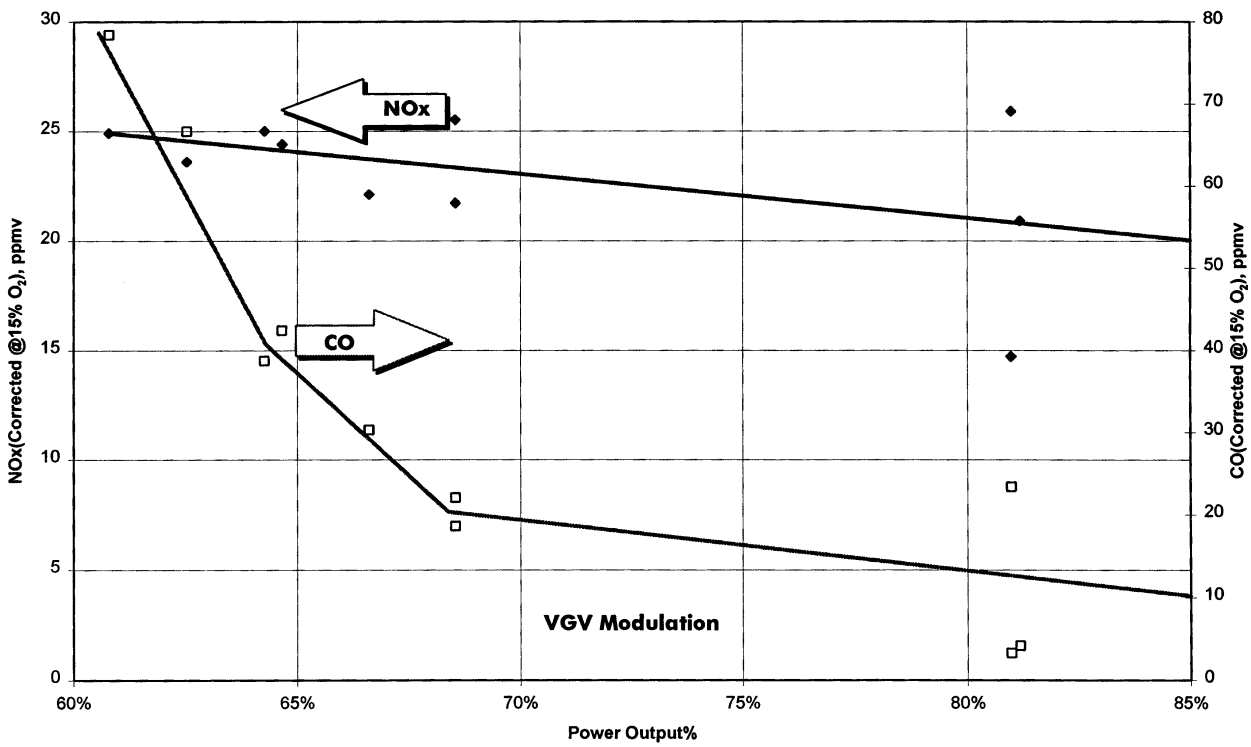

Fig. 11 The VGV modulation effect on CO depletion for the uprate Typhoon power range of 60–85 per cent

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more general terms:

$$\text{Wall temperature} = f\,(T_{\text{PRZ}}, m_{\text{cooling air}}, \text{impingement pressure drop and film cooling effectiveness})$$

The conventional film cooling technique, when employed in low NO_x lean burn combustors, produces high CO emissions. Early DLN development works proved the effect of combustor cooling on the depletion rate of the CO level. This is achieved through a series of emission traverse examination tests on the high-pressure air facilities. In order for CO specific targets to be met, it was recognized that film cooling was not an ideal solution. Utilization of the compressor air charged proportion to the casing through impingement holes subsequent to entering the combustion chamber was an attractive alternative.

An impingement cooling scheme was considered to deal with the can combustor cooling demands in order to spare more of the combustion air for premixing. The original design calculations were based on the analysis carried out by the author for an early DLN combustion system (i.e. G60). Figure 12 is a schematic of the impingement cooling scheme employed for the DLE can combustor. The one-dimensional analysis method was constructed and employed to evaluate the present DLE impingement combustor scheme and the effusion film creates heat transfer and cooling effectiveness. The film cooling devices evaluated were on selected areas of the combustion chamber and transition duct. The scheme achieved a lower reaction temperature and hence a reduction in the total NO_x emission. Also, it provided a hot wall boundary layer to avoid and/or reduce any wall quenching that could appear in the CO and UHC emissions. Moreover, the inner combustor surface was coated with a thin plasma-sprayed thermal barrier. The thermal barrier coating was introduced as an insulation layer between the hot gases and impingement cooled metal, thus introducing effectively a step change in the cooling effectiveness. It also allows a reduction in the coolant mass flow and/or its pressure loss for a specific design. The impingement air provided for the liner cooling was then sent through four discrete dilution holes. The dilution jets are utilized for the purpose of modulating the hot combustion gases pattern profile. The number of the impingement holes and the size were determined using modified Company in-house and Andrews and coworker correlations [**27**]:

$$\frac{x}{D} = c\left(\frac{\Delta P/P}{T_2}\right)^{c_1}\left(\frac{1}{G}\right)^{c_2}$$

The impingement liner side Nusselt number calculation was based on an experimental correlation that considers the effect of the streamwise and spanwise cross-flow and the resultant pressure drop. The calculation was carried out on the basis of a row-by-row steady state heat balance. The coolant side Nusselt number can be expressed as follows:

$$Nu = f\left(Re_{\text{jet}}, \left(\frac{x}{D}\right), \left(\frac{y}{D}\right), \left(\frac{z}{D}\right), \left(\frac{G_c}{G_j}\right)\right)$$

The predicted heat transfer value is a function of the local jet Reynolds number, cross-flow to jet mass velocity ratio and the impingement array geometry, i.e. the ratios of *x/D*, *y/D* and *z/D*. However, for the current DLE liners the average heat transfer coefficient was validated against Company in-house test data and Leeds University work.

A full one-dimensional model was constructed including correlations to calculate the radiation heat flux and gas heat flux due to forced convection for both gaseous and liquid fuels based on the latest information cited in reference [**28**]. The gas, material and fuel properties were

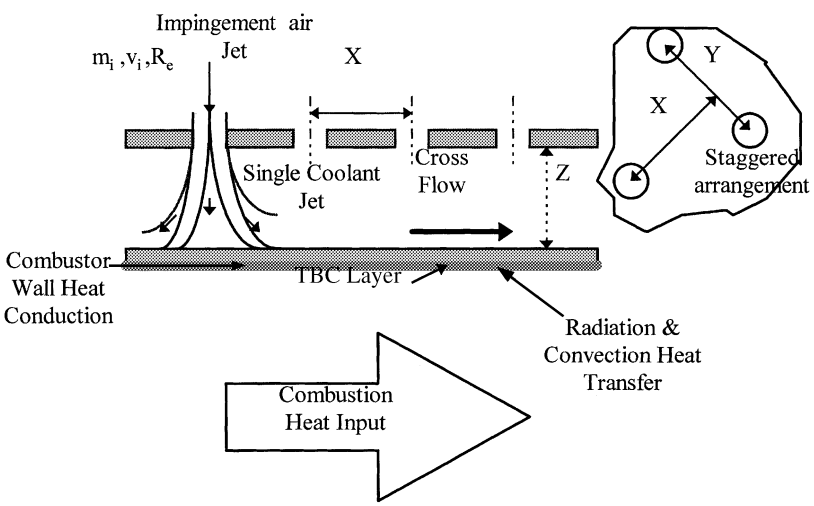

Fig. 12 Schematic representation of the current DLE impingement hole arrangements

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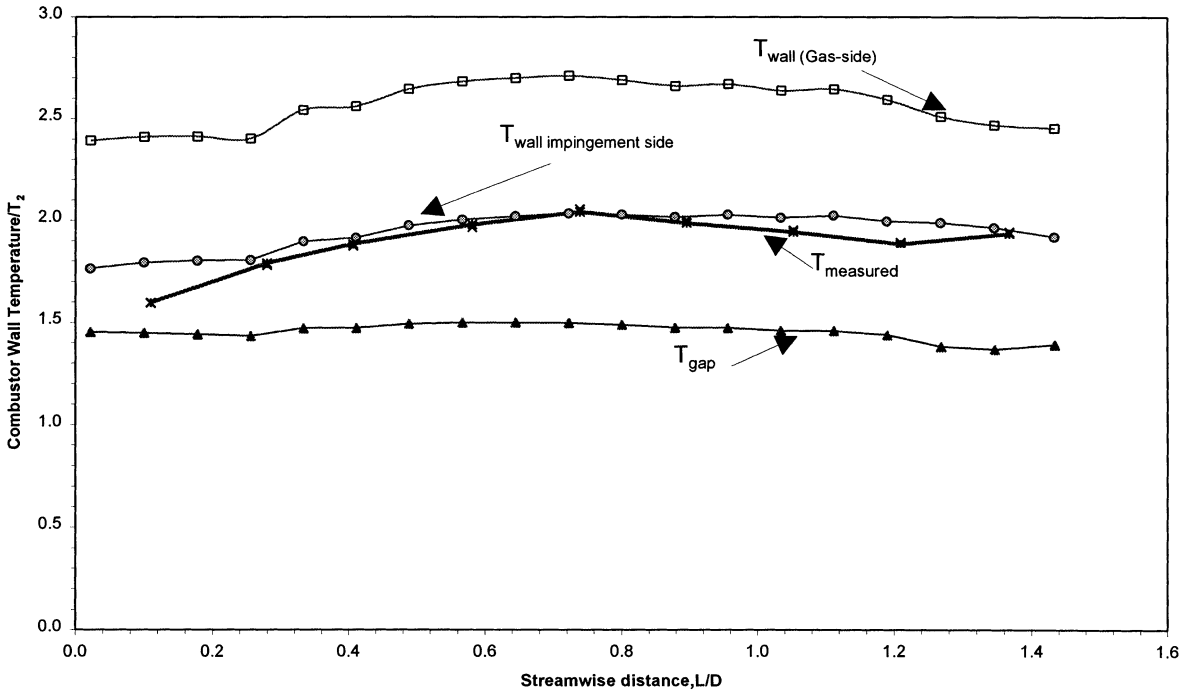

Fig. 13 Typical DLE combustor wall one-dimensional analysis as compared with the measured high-pressure test results

Company in-house database information. A reasonable agreement between the thermocouple's measured combustor wall temperatures and the calculated result is shown by Fig. 13 for a typical example of a DLE combustion system one-dimensional heat transfer analysis. The main difference in the calculated result was attributed to the assumption that the mean combustion temperature is even in the spanwise and streamwise directions for each row section. This could lead to an error, especially at the upstream region, due to the variation in the flame structure and flow turbulence intensity level from the mean value. The latter is considered a very important part in determining the film temperature near the wall region [**29**].

Future work will consider an improvement on the assumptions made for the one-dimensional calculation, especially that dealing with the effect of the near-combustor wall turbulent fluctuation in determining the surface heat transfer rate.

13 PATTERN FACTOR

The fuel and air non-uniformities may cause large-scale circumferential temperature maldistribution; thus the radial average temperature profile will be higher and consequently the blade life could be shorter. Recent development analyses and tests suggest that the geometrical size and location of transition piece cooling holes are crucial elements in the life of the hot parts downstream region of the combustion system. The large number of results from the high-pressure rig through the thermocouple probe and the production engine results through the pyrometer test all indicated a good practical pattern factor of $\leqslant 12$ per cent OTDF and $\leqslant 4$ per cent RTDF. This is attributed to adequate radial mixing and probably the enhancement obtained by the combined effect of the high mixing uniformity upstream region of the combustion chamber and the turbulent viscosity of the dilution and air bleeds jets from the transition duct. The size and location of the air bleed holes on the transition duct were optimized using a one-dimensional analysis method based on a modified correlation [**30**] to determine and evaluate the maximum air flow thermal penetration and optimal mixing. Several correlations can be used to estimate jet thermal mixing and penetration. Most studies that considered the penetration of dilution jets in cross-flow suggested that

$$\begin{pmatrix}\text{Jet thermal}\\ \text{mixing rate}\end{pmatrix} = f\begin{pmatrix}\text{momentum flux ratio, hole}\\ \text{spacing spanwise and}\\ \text{streamwise direction/}\\ \text{duct height}\end{pmatrix}$$

Figure 14a illustrates a typical example of one-dimensional analyses used to estimate air bleed jet thermal penetration. The transition duct effusion cooling holes affect the overall balance to give rise or fall to the cold outer or inner distribution of the temperature contour. Consequently, this will contribute to the radial

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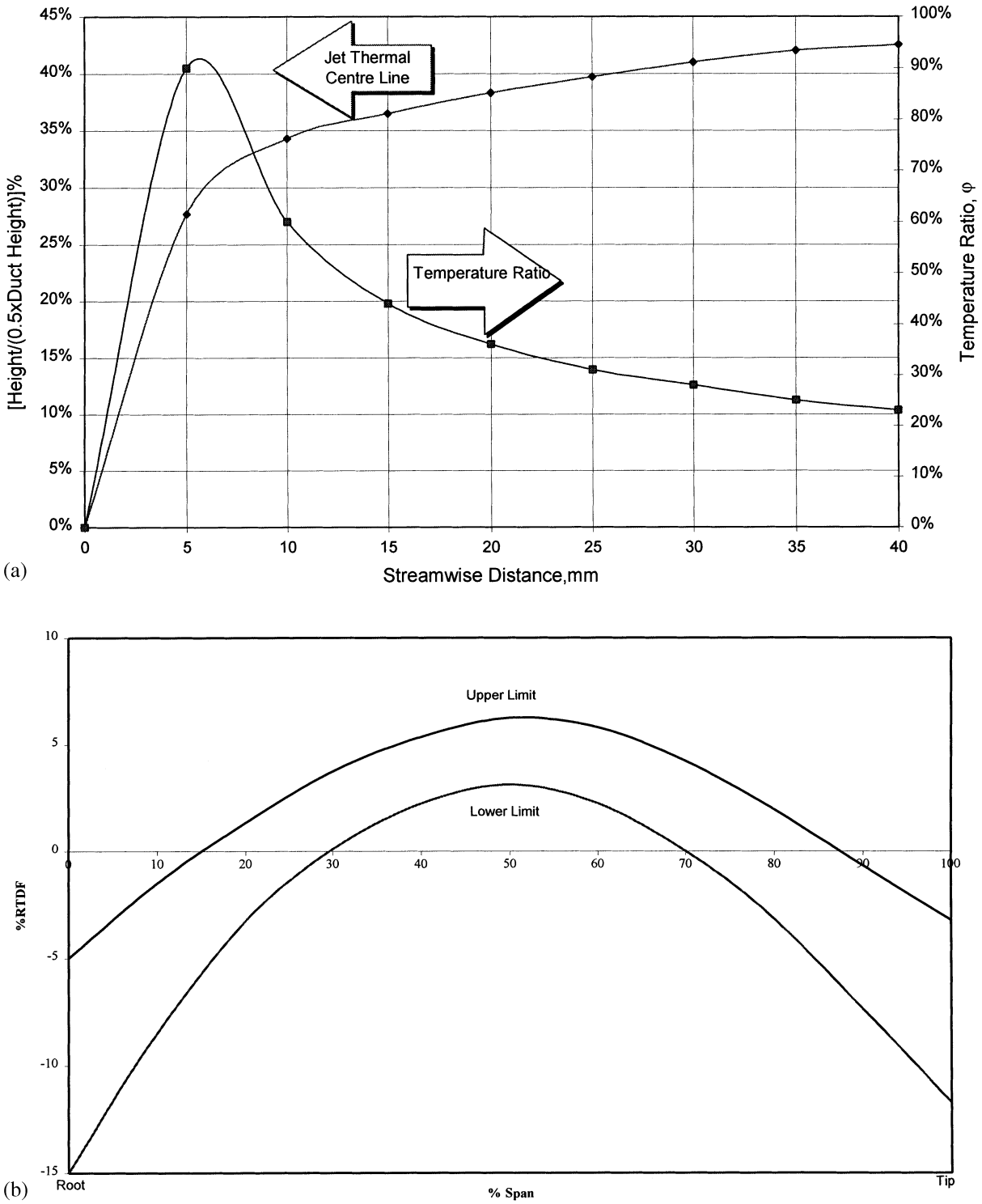

Fig. 14 (a) Typical dilution hole one-dimensional air jet thermal penetration analyses. (b) An illustrative example of the Company upper and lower RTDF% design limits

profile faced by the hot section downstream region of the transition duct, not so much the stationary but the rotating parts. Future work will publish the detailed analyses of the film cooling devices and their impact on the Company DLE combustion system.

The Company set up profile limits within practical tolerated and acceptable lower and higher radial temperature distribution factors (RTDF). The profile takes account of the field variability and component life. Figure 14b illustrates a typical RTDF design limitation con-

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straint that the combustion system pattern factor has to fulfil at the exit plane. The majority of test results using the pyrometer technique have demonstrated profiles that fall between these two limitations.

14 CFD SUPPORT FOR DLE COMBUSTION

Over the last few years, the use of the computer as an engineering tool has increased due to better computational software and facilities in terms of computer access, storage, speed and relative cost. There has also been a substantial increase in the use of computational fluid dynamics (CFD) in gas turbine combustor applications. It is important to note that the CFD applications for gas turbine combustors are driven not only by the potential cost and time savings but also by the ability to provide additional insight into complex problems that do not lend themselves to analytical solutions and may be too costly and time consuming to pursue experimentally.

The use of an effective tool to describe the internal flow field of the combustion system employed for ABB ALSTOM POWER engines was always considered part of its technology programme. The particular choice for a commercial CFD code to satisfy various criteria of the swirling combusting flow had proved to be difficult due to the many constraints regarding solver and/or methods limitation. STAR-CD CFD and other codes were employed to perform the necessary parametric comparisons for the combustion system.

A validation programme is currently undergoing rigorous consideration to find the means and methodology to improve CFD combusting flow analyses. Some feasibility studies involve the evaluation and comparison of two-dimensional with three-dimensional numerical methods. The saving would be in the scale of 1/24 h of CPU and man time. Figure 15 illustrates examples of the current CFD work carried out to give an insight into the current DLE combustor swirl flow field and temperature distribution. Figure 15a illustrates the STAR-CD numerical solutions for the velocity flow field ratio to the combustor mean approach velocity of the Tempest combustion system, while Fig. 15b demonstrates the two-dimensional simulation of the DLE combustor using CFX-Task flow. The latter code utilizes a turbulent flame speed based combustion model. The process is still undergoing considerable validation to prove its viability as a tool for the practical domain.

15 CONCLUSION

1. The full DLE burner assembly has demonstrated reasonable pressure loss and aerodynamics resistance to flashback to meet the constraints associated with the Company engines retrofit programme and the new design.
2. Fuel injection through the passages of the radial inflow swirler improves fuel and air mixing extensively. The bulk mean residence time was much less than the spontaneous ignition delay across the engine ranges. The partial premixing of the fuel and air aided by the delay in the vortex breakdown of the swirling flow, together with the aerodynamic shear layers, all contributed to establishing a well-stirred reactor.
3. The new method of a variable mixing regime utilizing the three zonal areas of radial passages, prechamber and the shear layer imposed by the swirler flow field has demonstrated a final low mixing parameter approaching the fully premixed system. One-dimensional analysis estimated the mean residence time of $\leqslant 2$ ms. This is the relaxation time needed for a nitrogen molecule to go from an excited to an equilibrium state and hence the reaction rate for NO_x formation at postcombustion temperatures must be very slow.
4. Better ignition characteristics and stability margins were achieved by the current pilot system. This is attributed to sufficient unmixedness generated by the pilot in the stabilizing swirling shear layer to give a considerable extension of the premixed stability limits imposed by the radial swirler. The fine-tuned programmed pilot schedule proved its adequacy across the engine power range. However, pilot fuel flow is contributing to the rise of NO_x emissions due to the deterioration of the local regional mixing parameter. It is paramount that the pilot/main swirler fuel schedule should utilize a rapid actuating pilot valve through fully integrated engine control system logic to accommodate any load range including the load shed condition.
5. The Company DLE combustion system exhibited low NO_x emissions of 15–20 ppmv (corrected to 15% O_2) across all engines ranges on development test beds and field trials. This is achieved with natural gas while operating a minimum pilot fuel split ratio at full-load conditions.
6. Ultra low NO_x emission $\leqslant 40$ ppmv (corrected to 15% O_2) is achieved by the ABB ALSTOM POWER DLE combustion system while operating at full thermal power with a minimum pilot fuel split using distillate-2 fuel.
7. The impingement cooling scheme aided by a thin layer of TBC was shown to give sufficient cooling effectiveness for the combustion chamber even though the coolant mass flowrate was reduced to $\leqslant 20$ per cent of the total mass delivered by the compressor to the combustion system. The maximum combustor wall temperature exhibited by the system was $\leqslant 800\,°C$ for various thermal loads and turbine entry temperatures.
8. The ABB ALSTOM POWER one-dimensional com-

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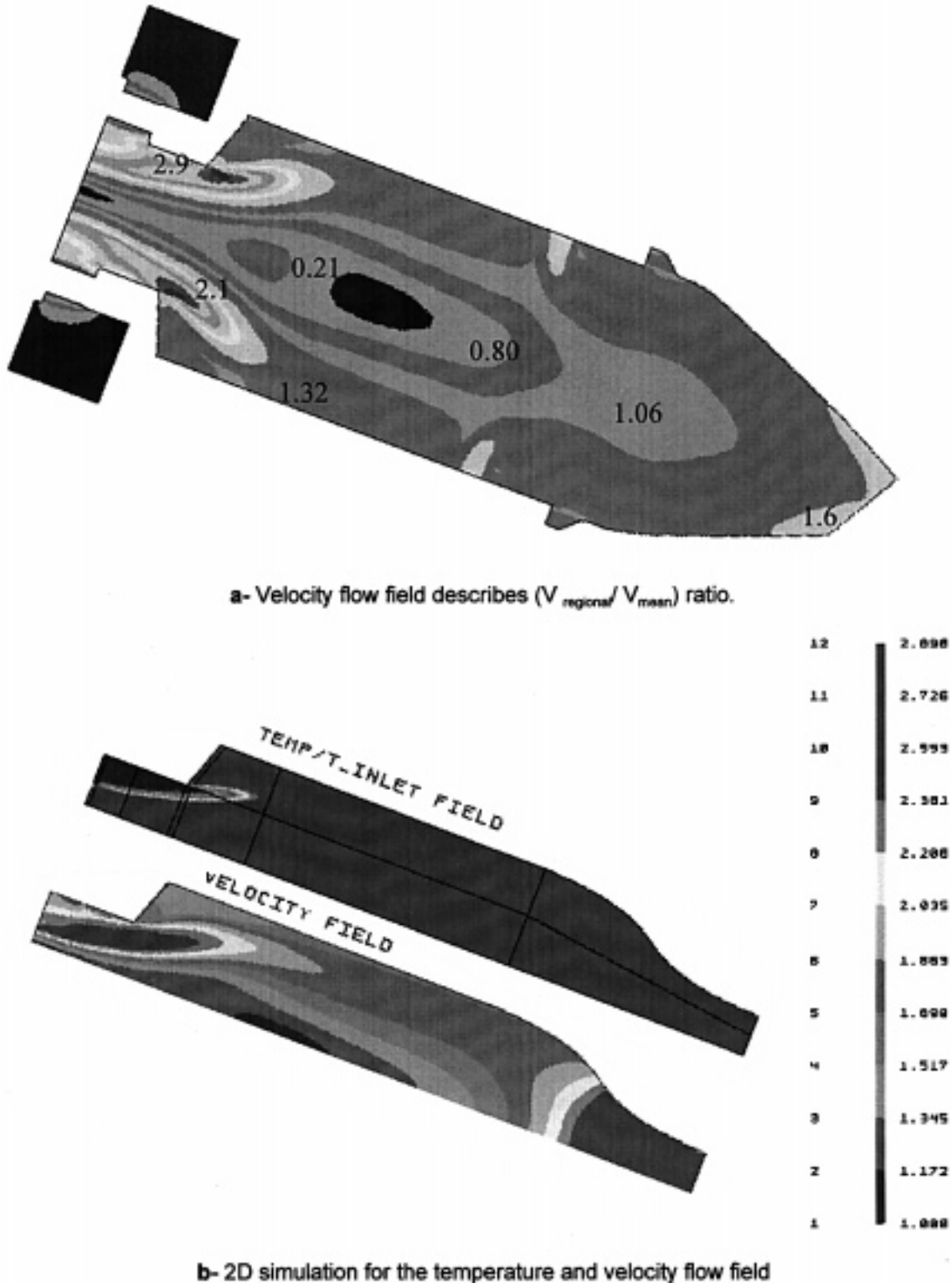

Fig. 15 Typical CFD results of the current DLE combustion system using three-dimensional and two-dimensional flow simulations

bustion design method is a useful tool to form preliminary design analyses and to characterize all its current DLE combustion system components and their integral parts. The analyses show good agreement between calculated and measured development test results.

9. The better air and fuel control management and the placement of the air split on the shield of the can

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combustor to match the local cooling requirement led to a design scheme that overcomes some harmful dynamics instabilities.

10. The turn-down ratio for the singles-shaft engine could be improved by closing the compressor variable guide vanes (VGV) to impose an optimum primary zone equivalence ratio of ~0.5 to allow for CO conversion to CO_2 without the penalty of increasing NO concentration or downgraded system performance.
11. The derived scaling factors for the Company DLE combustion system made it possible to read across its configuration and aerothermal performance to any power range without major conceptual changes. Also, it allows the dry solution to be used to control emissions where a wet solution could be out of reach unless power augmentation is available.
12. A balanced approach utilizing the fundamental one-dimensional analysis design method, experimental testing and two-/three-dimensional CFD-based modelling leads to the optimal development process.

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